

Decorators

A decorator modifies /extends the behavior of a function/method without changing the original code.

It's an external function that modifies its own behavior by modifying its internal function.

It will take another external function as an argument and will add it to its internal function.

A function that takes another fucntion as an argument(without a decorator)

An outer function "A" that uses "C" to modify "B"

- 1. with a "B" internal function that will be modified

Another outer function "C"

- 1. that is used as an argument by "A" and that will be used to modify the behavior of "B".

General Example below

```
1 def A(func):
2     def B():
3         print("First")
4         func() # Here "C" will be executed
5         print("Second")
6     return B
7
8 #-----NO DECORATOR is bein used here
9 def C():
10     print("Hello")
11
12 c_hello= A(C) # Here we instantiate an object of the function A using teh variable c_hello
13 c_hello()     # Execution of c_hello
```

First
Hello
Second

A function that takes another fucntion as an argument USING a DECORATOR

```
1
2 def A(func):
3     def B():
4         print("First")
5         func() # Here "C" will be executed
6         print("Second")
7     return B
8
9 @A # Here we use the use of a DECORATOR by using the label @ to indicate that the below function will be takend as argument
10 def C():
11     print("Third")
12
13 C() # No need to instanatiate an object of the function. Just execute/call
14
15
```

First
Third
Second

```
1
2
3 def decorator(func): #-----The parameter (func) will be replaced by the function ( bite() )
4     def dog(): # This is teh inner function
5         print("I can bark but I cannot bite")
6         func() #-----This adds biting capability. So (func) parameter is replaced by a function bite() so now is func = func()
7         print("Now I can bark AND bite")
8     return dog # When the function 'dog' does not have the parentheses() it means is WAITING to be called/executed
9
10 @decorator #This line is used to take the bite() function as a parameter for the decorator function
11 def bite():
12     print("I can bite")
13
14 bite() # Call/Execute the function executes teh whole process
15
16
```

I can bark but I cannot bite
I can bite
Now I can bark AND bite